MAE 280B
Linear Control Design – Winter 2013

Midterm

Instructions:

- Due on 02/15/2013 at my office (EBU I 1602) by 5:00 PM;
- Use Matlab or Mathematica;
- You get marks for clarity;
- You loose marks for obscurantism;
- This exam has 4 questions, for a total of 33 points and 9 bonus points.

1. (4 points) **Quadratic Matrix Equation.** Show that there exists a matrix K that solves the matrix equation

$$(A + BK)^T X + X(A + BK) + (C + DK)^T W\,(C + DK) = 0$$

if and only if there exist a matrix $\tilde{K}$ that solves the matrix equation

$$(\tilde{A} + B\tilde{K})^T X + X(\tilde{A} + B\tilde{K}) + \tilde{Q} + \tilde{K}^T D^T W\, D\, \tilde{K} = 0$$

where

$$\tilde{A} = A - B(D^T W D)^{-1} D^T W C, \qquad \tilde{Q} = C^T [W - WD(D^T W D)^{-1} D^T W] C.$$

Furthermore, if a solution exists,

$$K = \tilde{K} - (D^T W D)^{-1} D^T W C.$$

Hint: Complete the squares in K.

2. **Removing Assumptions on State Estimation.** Consider the LTI system

$$\begin{aligned} \dot{x} &= Ax + B_w w, \qquad x(0) = 0, \\ y &= C_y x + D_{yw} w, \end{aligned} \tag{1}$$

where w is a Gaussian zero-mean white noise with covariance matrix $W \succ 0$, (A, C_y) is observable, and the state observer

$$\begin{aligned} \dot{\hat{x}} &= A\hat{x} + F(\hat{y} - y), \qquad \hat{x}(0) = 0, \\ \hat{y} &= C_y \hat{x}. \end{aligned}$$

(a) (2 points) Show how to compute the cost function

$$J = \lim_{t \to \infty} E\left[e(t)^T e(t)\right], \qquad e(t) = x(t) - \hat{x}(t) \tag{2}$$

in terms of Gramians. Compute both primal (observability) and dual (controllability) Gramian formulas and the associated Lyapunov equations.

(b) (4 points) Assuming $B_w W D_{yw}^T \neq 0$ and $D_{yw} W D_{yw}^T \succ 0$, show all the steps needed to calculate a state estimator gain F that minimizes J.

Hint: Use Question 1 to handle $B_w W D_{yw}^T \neq 0$. You do not need to prove anything that we already proved in class but you will need to state the assumptions under which your calculations hold.

3. **Removing Assumptions on LQG Control.** Consider the LTI system

$$
\begin{aligned}
\dot{x} &= Ax + B_w w + B_u u, \qquad x(0) = 0,\\
y &= C_y x + D_{yw} w + D_{yu} u,\\
z &= C_z x + D_{zw} w + D_{zu} u,
\end{aligned} \tag{3}
$$

where w is a Gaussian zero-mean white noise with covariance matrix $W \succ 0$, (A, B_u) is controllable, and (A, C_y) is observable.

(a) (1 point (bonus)) Show that the cost function

$$J = \lim_{t\to\infty} E\left[z(t)^T z(t)\right]. \tag{4}$$

is finite only if $D_{zw} = 0$.

Hint: Compute the transfer-function from w to z and use Parseval's Theorem.

On the next questions assume $D_{zw} = 0$.

(b) (4 points) Assuming $C_z^T D_{zu} = 0$, $B_w W D_{yw}^T = 0$, $D_{zu}^T D_{zu} \succ 0$, $D_{yw} W D_{yw}^T \succ 0$, compute the state-space realization and the transfer-function of the optimal LQG controller which minimizes the cost function (4) for the system

$$
\begin{aligned}
\dot{x} &= Ax + B_w w + B_u u, \qquad x(0) = 0,\\
\tilde{y} &= C_y x + D_{yw} w,\\
z &= C_z x + D_{zu} u,
\end{aligned} \tag{5}
$$

where w is a Gaussian zero-mean white noise with covariance matrix $W \succ 0$, (A, B_u) is controllable, and (A, C_y) is observable.

Hint: You can use results from the lecture notes if you state the assumptions needed for them to hold.

(c) (4 points) Systems (3) and (5) are related through:

$$\tilde{y} = y - D_{yu} u.$$

Let $C(s)$ be the controller you computed in part (b). Show that

$$C(s) = [I + \tilde{C}(s) D_{yu}]^{-1} \tilde{C}(s)$$

is the optimal LQG controller for the original system (3).

(d) (4 points (bonus)) Repeat part (b) assuming $B_w W D_{yw}^T \neq 0$, $B_w W D_{yw}^T \succ 0$.

Hint: Use Question 2.

4. **Control Design.** Consider the system

$$G(s) = \frac{s^2}{s^2 - 1}. \tag{6}$$

(a) (6 points) Using any "classical" control technique (root-locus, bode diagram, Nyquist criterion, etc) find a controller that can stabilize $G(s)$. What is the order of the controller you found? Is the controller itself stable? Can you explain why?

Hint: Do not cancel zeros or poles on the right hand side of the complex plane!

(b) (1 point) Compute a state space representation (A, B_u, C_y, D_{yu}) for $G(s)$.

(c) (4 points) With Question 3 in mind, let the input $u(t)$ be perturbed by $w(t)$ and the measurement output $\tilde{y}(t)$ be perturbed by $v(t)$ as in:

$$\begin{aligned} \dot{x} &= Ax + B_u(u + w) \\ \tilde{y} &= C_y x + v. \end{aligned}$$

Assuming that $w(t)$ and $v(t)$ are independent Gaussian zero-mean white-noise with unitary variance, find optimal LQG controllers $\tilde{C}(s)$ that minimize the cost function

$$J = \lim_{t\to\infty} E\left[z(t)^T z(t)\right], \qquad z = \begin{pmatrix} \tilde{y} - v \\ \rho u \end{pmatrix}$$

for choices of $\rho = \{10^{-6}, 10^{-3}, 1, 10^3, 10^6\}$. Compute the corresponding controller $C(s)$ using Question 3(c).

(d) (4 points) What is the impact of the different choices of ρ on the cost function, the closed-loop eigenvalues, the controller poles, the controller zeros? Is any of these controllers stable? Do you think you can find a stable stabilizing controller? Why?

IMPORTANT: Answer the items related to the controller with respect to $C(s)$, that is the controller of the original system with transfer function $G(s)$, not $\tilde{C}(s)$!

(e) (4 points (bonus)) Let the input $u(t)$ be perturbed by $w(t)$ and the measurement output $y(t)$ be perturbed by $v(t)$ as in:

$$\begin{aligned} \dot{x} &= Ax + B_u(u + w) \\ y &= C_y x + D_{yu}(u + w) + v. \end{aligned}$$

Assuming that $w(t)$ and $v(t)$ are independent Gaussian zero-mean white-noise with unitary variance, find optimal LQG controllers $C(s)$ that minimize the cost function

$$J = \lim_{t\to\infty} E\left[z(t)^T z(t)\right], \qquad z = \begin{pmatrix} y - D_{yu}w - v \\ \rho u \end{pmatrix}$$

for choices of $\rho = \{10^{-6}, 10^{-3}, 1, 10^3, 10^6\}$.

Hint: Use Question 3.